In [5]:

```python
import numpy as np

# Input patterns
patterns = np.array([


    [1, 0, 1, 0],
    [1, 1, 0, 0],
    [0, 0, 1, 1],
    [0, 1, 0, 1]
])

vigilance = 0.5
clusters = []

for p in patterns:
    matched = False

    for c in clusters:
        similarity = np.sum(p & c) / np.sum(p)

        if similarity >= vigilance:
            matched = True
            break

    if not matched:
        clusters.append(p)

print("Number of clusters formed:", len(clusters))
```

```
Number of clusters formed: 2
```

In [ ]:

In [ ]: